

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Botany**Paper: BOB-502 — Comparative Studies of Phanerogams**

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions in all, selecting two each from Sections A and B and Question 9 (Section-C) which is compulsory. The figures in the right-hand margin indicate marks.

SECTION A

Question 1. Give a detailed account of reproduction in *Gnetum*. [15]

Question 2. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Distribution of gymnosperms in India
- (b) *Pentoxylon*
- (c) Male and female cones of *Biota*
- (d) Embryogeny in *Ginkgo*

Question 3. Give a brief account of morphology and reproduction in *Zamia*. [15]

Question 4. Summarize phylogenetic trends in gymnosperms. [15]

SECTION B

Question 5. Discuss Hutchinson's classification, its merits and demerits. [15]

Question 6. Discuss briefly any two of the following: [2 × 7.5 = 15]

- (a) Numerical taxonomy
- (b) Embryo culture
- (c) Parthenogenesis
- (d) Cleavage polyembryony

Question 7. Write salient features of the family Euphorbiaceae and discuss its economic importance. [15]

Question 8. Differentiate between any three of the following: [3 × 5 = 15]

- (a) Diplospory and Apospory

- (b) Ray and Disc florets
- (c) Scrophulariaceae and Verbenaceae
- (d) Zingiberaceae and Liliaceae

SECTION C

Question 9. Answer each of the following within 50 words: [10 × 1 = 10]

- (a) Name any monocot plant whose leaves do not show parallel venation.
- (b) Write about hesperidium.
- (c) Name the scientists who first reported the development of plant from anther-culture.
- (d) Mention any two resemblances of *Gnetum* leaf with that of angiosperms.
- (e) Who reconstructed the genus *Williamsonia seawardiana*?
- (f) What do you mean by syndetocheilic stomata?
- (g) Mention peculiar features of *Cuscuta* embryo.
- (h) Give botanical names of any two medicinal plants associated with the Combretaceae family.
- (i) Write about Cyathium inflorescence.
- (j) What is Ochrea?

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